

Twin Turbo: Hybrid Edge Intelligence and Digital Twins for Autonomous Turbofan Maintenance

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Abstract—Deciding when to remove a jet engine from service trades the cost of an in-service failure against the value of the life left unused. Deep-learning models for remaining-useful-life (RUL) prediction report steadily lower error on the NASA C-MAPSS benchmark, but they output point estimates without calibrated confidence, and they are evaluated on prediction error rather than on the maintenance decisions they would drive. This paper presents *Twin Turbo*, a framework that combines a lightweight edge model with a per-engine digital twin and makes autonomous maintenance decisions from calibrated lower bounds. The edge model is a gradient-boosted tree ensemble over 30-cycle sensor windows, wrapped in Mondrian conformal prediction so that its intervals carry a distribution-free coverage guarantee. The digital twin is a particle filter over an exponential health-index degradation model: a population prior learned from run-to-failure engines is updated with the observations of one specific engine, yielding a full RUL distribution. The edge model runs every cycle; the twin is synchronized only when one of three triggers fires, and the policy acts on the more cautious of the two calibrated bounds. On the official C-MAPSS test sets the twin reaches an RMSE of 10.33 on FD002, and its 90% intervals achieve 86–94% empirical coverage, against 43–54% for Monte-Carlo dropout on an LSTM baseline. Replaying 709 held-out engines from first cycle to failure through five-fold cross-validation, the hybrid policy lets no engine fail, lowers the long-run maintenance cost rate by 34–47% relative to optimal age replacement, consults the twin on only 21–28% of cycles, and is the only policy evaluated that remains failure-free without any safety buffer on all four datasets.

Index Terms—prognostics and health management, remaining useful life, digital twin, particle filter, conformal prediction, edge computing, predictive maintenance, C-MAPSS

I. INTRODUCTION

Maintenance accounts for a large share of the operating cost of aircraft engines, and the central decision is deceptively simple: when should an engine come off the wing? Removing it too late risks an unplanned failure in service, the most expensive and most dangerous outcome. Removing it too early discards flight cycles that were already paid for. The classical answer is *time-based* maintenance: retire every unit at a fixed age chosen from fleet lifetime statistics [1]. It is safe on average but wasteful, because it treats the longest-lived engine exactly like the shortest-lived one. *Condition-based* maintenance instead uses each engine’s own sensor data to estimate its remaining useful life (RUL) and act on it [2], [3].

The NASA C-MAPSS dataset [4], which simulates fleets of turbofans degrading from a healthy state to failure under noise, multiple operating regimes and multiple fault modes, has become the standard benchmark for RUL estimation [5]. Over the last decade, reported accuracy on it has improved steadily through convolutional [6], [7], recurrent [8], hybrid [9] and attention-based [10] networks. Three gaps nevertheless separate these models from an autonomous maintenance system.

First, point estimates are not decisions. A model that predicts 40 cycles of remaining life with an error of ± 5 cycles supports a very different action from one that predicts 40 cycles with an error of ± 40 . Most published C-MAPSS models output a single number. Where uncertainty is added, typically through Monte-Carlo dropout [11], its calibration is rarely checked, and benchmarks of uncertainty methods for prognostics show that calibration quality varies widely [12].

Second, accuracy is measured in the wrong place. The benchmark protocol scores each test engine only at the last cycle of a truncated trajectory. This reveals nothing about how a model behaves as an engine approaches failure, whether a policy built on it would ground engines in time, or what that policy would cost relative to simply retiring engines at a fixed age.

Third, “digital twin” is used loosely. The concept, as articulated for aerospace by Glaessgen and Stargel [13] and formalized by Grieves and Vickers [14], denotes a virtual model bound to one specific physical asset and kept synchronized with it. Neural RUL regressors trained on a whole fleet hold no per-asset state and are not synchronized with anything, yet they are frequently presented as twins.

At the same time, practical deployment imposes a systems constraint. A model that runs on or near the engine every cycle must be small and stateless, while a model that maintains a detailed belief about each engine is naturally hosted off board, where consulting it costs communication and connectivity. Edge computing [15] and cascaded inference [16] suggest the obvious pattern: run the cheap model always, and escalate to the expensive one only when needed.

This paper presents *Twin Turbo*, a framework that addresses all three gaps within that edge/twin pattern. A gradient-boosted edge model runs on every cycle and is wrapped in

Mondrian conformal prediction [17], [18], giving it calibrated intervals at negligible cost. A digital twin, instantiated once per engine, tracks a sensor-fused health index [19] with a particle filter [20], [21] over an exponential degradation model whose population prior is learned from run-to-failure engines [22]. The twin is synchronized only when the edge model raises one of three triggers (low predicted RUL, a sharp one-cycle drop, or a periodic resynchronization deadline), and a fixed policy maps the more cautious of the two calibrated lower bounds to one of four actions: continue, watch, schedule maintenance, or ground now.

The contributions of this work are:

- 1) **An instance-specific, synchronizable digital twin** for turbofan prognostics, combining a learned health index with Bayesian particle filtering of a fleet prior. On the official C-MAPSS test sets it attains RMSE between 10.33 and 14.78, and its 90% posterior intervals are empirically calibrated (86–94% coverage) without tuning.
- 2) **A hybrid edge/twin decision architecture** with conformally calibrated edge intervals, trigger-based twin synchronization, and a cautious fusion rule that preserves at least 90% coverage by the union bound.
- 3) **A decision-level evaluation protocol** that replays 709 held-out engines from first cycle to failure and scores every policy by failures, unused life and long-run renewal-reward cost rate, against optimal age replacement [1], edge-only, twin-only, LSTM and oracle baselines, with ablations of every design choice. Under this protocol the hybrid policy prevents every failure, reduces cost rate by 34–47% relative to age replacement, and consults the twin on 21–28% of cycles.
- 4) **An open, reproducible implementation** with an interactive console that replays engines, the fleet and the particle filter live from the same traces used for the reported results.

The remainder of the paper is organized as follows. Section II surveys prognostics methods, uncertainty quantification, digital twins, edge inference and maintenance decision theory, and positions this work among them. Subsequent sections describe the dataset and preprocessing, the edge model, the digital twin and the decision layer, followed by the experimental protocol, results, ablations and a discussion of limitations.

II. LITERATURE SURVEY

A. Prognostics and the C-MAPSS benchmark

Prognostics estimates the *remaining useful life* (RUL) of an asset from condition-monitoring data, so that maintenance can be planned before failure rather than after it [2], [3]. Approaches are usually grouped into physics-based models, statistical (data-driven stochastic) models, and machine-learning models [3], [23]. The C-MAPSS dataset [4] was released by NASA for the 2008 PHM data challenge: turbofan engines are simulated from a healthy state to failure under randomized initial wear, sensor noise, one or six operating regimes and one or two fault modes. It remains the most widely used RUL

TABLE I
PUBLISHED TEST-SET RMSE ON C-MAPSS FOR REPRESENTATIVE METHODS, AS COMPILED BY RAGAB ET AL. [10]. LOWER IS BETTER.

Method	FD001	FD002	FD003	FD004
Gradient boosting [28]	15.67	29.09	16.84	29.01
2-D CNN [6]	18.45	30.29	19.82	29.16
Deep LSTM [8]	16.14	24.49	16.81	28.17
1-D deep CNN [7]	12.61	22.36	12.64	23.31
MODBNE ensemble [28]	15.04	25.05	12.51	28.66
BiLSTM enc.-dec. [27]	14.74	22.07	17.48	23.49
Hybrid DNN [9]	13.02	15.24	12.22	18.16
Attention seq2seq [10]	12.63	14.65	11.44	16.66

Figures for the same method can differ slightly between compilations (e.g. 16.18 rather than 16.81 is also reported for the deep LSTM on FD003); we use one source throughout for consistency.

benchmark; Ramasso and Saxena [5] surveyed the methods applied to it and documented pitfalls in how results are reported, including inconsistent RUL clipping and evaluation protocols.

B. Statistical and model-based prognostics

Two strands from this literature directly shape our digital twin. *Health-index (HI) construction*: Wang et al. [19] (a winning approach at PHM08) fuse sensors into a single HI by regressing sensor values onto 1 for early-life and 0 for end-of-life cycles, then match a test engine’s HI curve against a library of training trajectories. *Stochastic degradation models with Bayesian updating*: Gebraeel et al. [22] model degradation as an exponential process whose parameters have a population prior learned from a fleet, and update that prior with each new observation of the individual unit to obtain a residual-life *distribution*. Particle filters [20] generalize this updating to nonlinear, non-Gaussian settings; Orchard and Vachtsevanos [21] established the particle-filtering framework for online fault diagnosis and failure prognosis, and the Liu–West kernel-shrinkage scheme [24] is the standard way to estimate static model parameters inside a particle filter without particle collapse. Such models are interpretable and yield full predictive distributions, but are seldom benchmarked against modern deep models on C-MAPSS.

C. Deep learning for RUL

Since 2016, deep networks trained on sliding windows of raw sensor data have dominated reported C-MAPSS accuracy. Babu et al. [6] introduced convolutional networks for RUL regression; Zheng et al. [8] applied LSTMs [25]; Li et al. [7] showed that deeper 1-D convolutions with a piecewise-linear RUL target reach RMSE near 12.6 on FD001. Later work added semi-supervised pre-training [26], encoder–decoders [27], hybrid CNN–LSTM architectures [9] and attention [10]. Table I compiles representative results. Two gaps are consistent across this line of work: (i) models output a point estimate, with no calibrated measure of confidence, and (ii) accuracy is reported only at the last observed cycle of each truncated test engine, which says nothing about whether the resulting *maintenance decisions* are safe or economical.

Gradient-boosted trees [29], [30] on hand-crafted window features remain a strong, far cheaper baseline.

D. Uncertainty quantification

Monte-Carlo dropout [11] and deep ensembles [31] are the most common ways to attach uncertainty to neural RUL predictors. Their calibration is not guaranteed: Basora et al. [12] benchmark these methods for prognostics and find that calibration quality varies substantially across methods and engines. Conformal prediction [17], [18] instead wraps any point predictor and provides intervals with a finite-sample, distribution-free coverage guarantee under exchangeability; Mondrian (group-conditional) variants calibrate separately within bins, which matters for RUL where error grows with distance from failure. To our knowledge conformal intervals are still rarely used to drive maintenance actions on C-MAPSS.

E. Digital twins

The digital-twin concept was articulated for aerospace by Glaessgen and Stargel [13] as a model that “mirrors the life of its flying twin” by combining physics, fleet history and the vehicle’s own sensor data, and was formalized by Grieves and Vickers [14] as a virtual counterpart linked to one specific physical instance and kept synchronized with it. Tao et al. [32] review industrial uses, with prognostics and health management among the most prominent. Two properties recur in these definitions and are often missing from works that label a neural RUL regressor a “digital twin”: the model is *instance-specific* (it holds state about one asset) and it is *synchronized* with that asset as data arrive.

F. Edge intelligence and cascaded inference

Edge computing places computation near data sources to reduce latency, bandwidth and dependence on connectivity [15]. Cascaded or early-exit inference [16] runs a cheap model on every input and defers to an expensive one only when the cheap model is not confident enough. Our framework applies the same idea across a device/twin boundary: a stateless edge model runs every cycle, and the stateful twin is synchronized only when a trigger fires.

G. From predictions to maintenance decisions

Classical reliability theory gives the baseline any predictive policy must beat: *age replacement*, which replaces a unit at a fixed age chosen to minimize the long-run cost rate of the renewal process [1]. We evaluate every policy with the same renewal-reward cost rate, so predictive maintenance is compared against this optimal time-based policy and not only against run-to-failure.

H. Positioning of this work

Existing C-MAPSS studies optimize last-cycle RMSE of point predictors; existing digital-twin work is mostly conceptual or not benchmarked on it. This project combines (i) an instance-specific, Bayesian-updated degradation twin in the tradition of [21], [22], (ii) a conformally calibrated edge model, and (iii) a trigger-based synchronization scheme,

TABLE II
TYPICAL PROPERTIES OF EACH FAMILY OF APPROACHES AS APPLIED TO C-MAPSS, AND OF THIS WORK.

Approach	Calibrated uncertainty	Per-asset state	Decisions evaluated
Similarity-based HI matching [19]	–	–	–
Bayesian degradation / particle filters [21], [22]	posterior	✓	rarely
Deep point regressors [6]–[8], [10]	–	–	–
Deep nets + MC dropout / ensembles [11], [31]	not guaranteed [12]	–	rarely
Conformal prediction [17], [18]	✓	–	rarely
Age replacement [1]	n/a	–	✓
This work	✓	✓	✓

and evaluates the combination on what matters operationally: failures prevented, useful life used and cost rate, over engines replayed all the way to failure. Table II summarizes how this combination relates to the families surveyed above.

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